

# MCD 4710 Assignment 2

## –

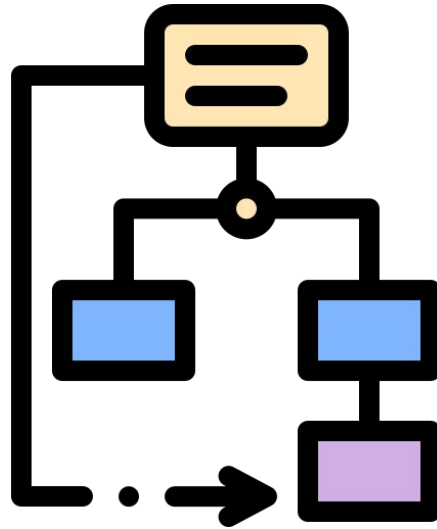
# Milestone 1

Team 3: Alexander, Dyllon,  
Nathan, Angad

Google slides link:

<https://docs.google.com/presentation/d/1vJPga0flw9ExtK3SpsUNqfn-UDItlCsNdZrA8ORjEZ0/edit?usp=sharing>

# UML Diagrams

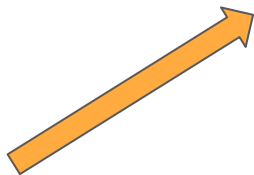


## Base Class -

| LibraryItem (BASE)                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>-item_id: str</li><li>-title: str</li><li>-status: str (Available / borrowed)</li><li>-type: str</li></ul>                                                                                                                                                                                                                                                                                                                             |
| <ul style="list-style-type: none"><li>+__init__(item_id: str, title: str, type: str, status: str = "Available")</li><li>+get_item_id() :str</li><li>+get_title() : str</li><li>+get_status() : str</li><li>+get_type() : str</li><li>+get_borrow_duration(): int</li><li>+set_item_id(new_item_id: str): void</li><li>+set_title(new_title: str): void</li><li>+set_status(new_status: str): void</li><li>+set_type (new_type: str): void</li><li>+__str__() → str</li></ul> |

# Base Class (LibraryItem)

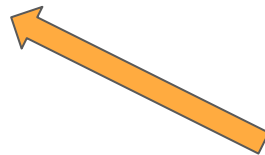
Derived Class



Derived Class



Derived Class



PrintedBook

- author: str  
- number\_of\_pages: int

+ \_\_init\_\_(item\_id: str, title: str, author: str,  
number\_of\_pages: int, status: str = "Available")

+get\_author(): str  
+get\_number\_of\_pages(): int  
+get\_borrow\_duration(): int (returns 14)

+set\_author(new\_author: str)  
+set\_number\_of\_pages(pages: int)

+ \_\_str\_\_() : str

EBook

- file\_size: str  
- format: str

+ \_\_init\_\_(item\_id: str, title: str, file\_size: str, format: str,  
status: str = "Available")

+get\_file\_size(): str  
+get\_format(): str  
+get\_borrow\_duration(): int (returns 7)

+set\_file\_size (new\_file\_size: str): void  
+ set\_format (new\_format : str): void

+ \_\_str\_\_(): str

Magazine

- issue\_number: str  
- publication\_date: str

+ \_\_init\_\_(item\_id: str, title: str, issue\_number: str,  
publication\_date: str, status: str = "Available")

+get\_issue\_number(): int  
+get\_publication\_date(): str  
+get\_borrow\_duration(): int (returns 3)

+set\_issue\_number(issue: str)  
+set\_publication\_date(date: str)

+ \_\_str\_\_() → str

## Member

-member\_name : str  
-contact\_number: str  
-borrowed\_items\_list: list

+ \_\_init\_\_(member\_name: str, contact\_number: str)  
  
+ register(): Member  
+ get\_member\_name(): str  
+ get\_contact\_number(): str  
+ get\_borrowed\_items\_list(): list  
  
+set\_contact\_number(new\_contact: str): void  
+set\_member\_name(new\_name:str): void  
  
+ borrow\_item(item: LibraryItem): bool  
+ return\_item(item: LibraryItem): bool  
  
+ \_\_str\_\_(): str

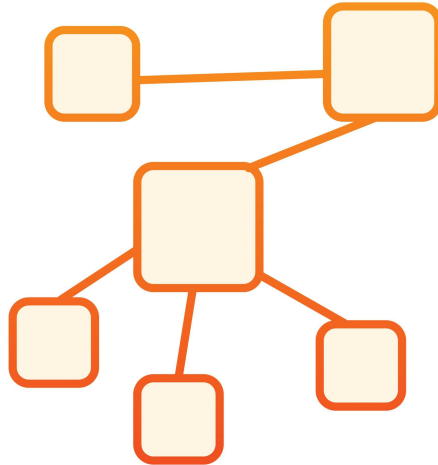
## BorrowRecord

-member : Member  
-item: Library\_item  
-borrow\_date: str  
-return\_date: str

+ \_\_init\_\_(member: Member, item: LibraryItem, borrow\_date: str, return\_date: str  
= None)  
  
+ get\_member(): Member  
+ get\_item(): LibraryItem  
+ get\_borrow\_date(): str  
+ get\_return\_date(): str  
  
+ set\_member(member: Member): void  
+ set\_item(item: LibraryItem): void  
+ set\_borrow\_date(date: str): void  
+ set\_return\_date(date: str): void  
  
+ \_\_str\_\_(): str

| Staff                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>-staff_id: str</p> <p>-staff_name: str</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p>+__init__(staff_id: str, staff_name: str)</p> <p>+get_staff_id() : str</p> <p>+get_staff_name() : str</p> <p>+set_staff_id(new_staff_id:str: str) : void</p> <p>+set_staff_name(new_staff_name: str): void</p> <p>+add_item(item_list: list, item: LibraryItem): bool</p> <p>+edit_item(item_list: list, item: LibraryItem): bool</p> <p>+delete_item(item_list: list, item: LibraryItem): bool</p> <p>+view_all_items(item_list: list): list</p> <p>+save_data(item_list: list, filename:str): bool</p> <p>+__str__(): str</p> |

# Class Diagram



(Made using Lucidchart)

link:

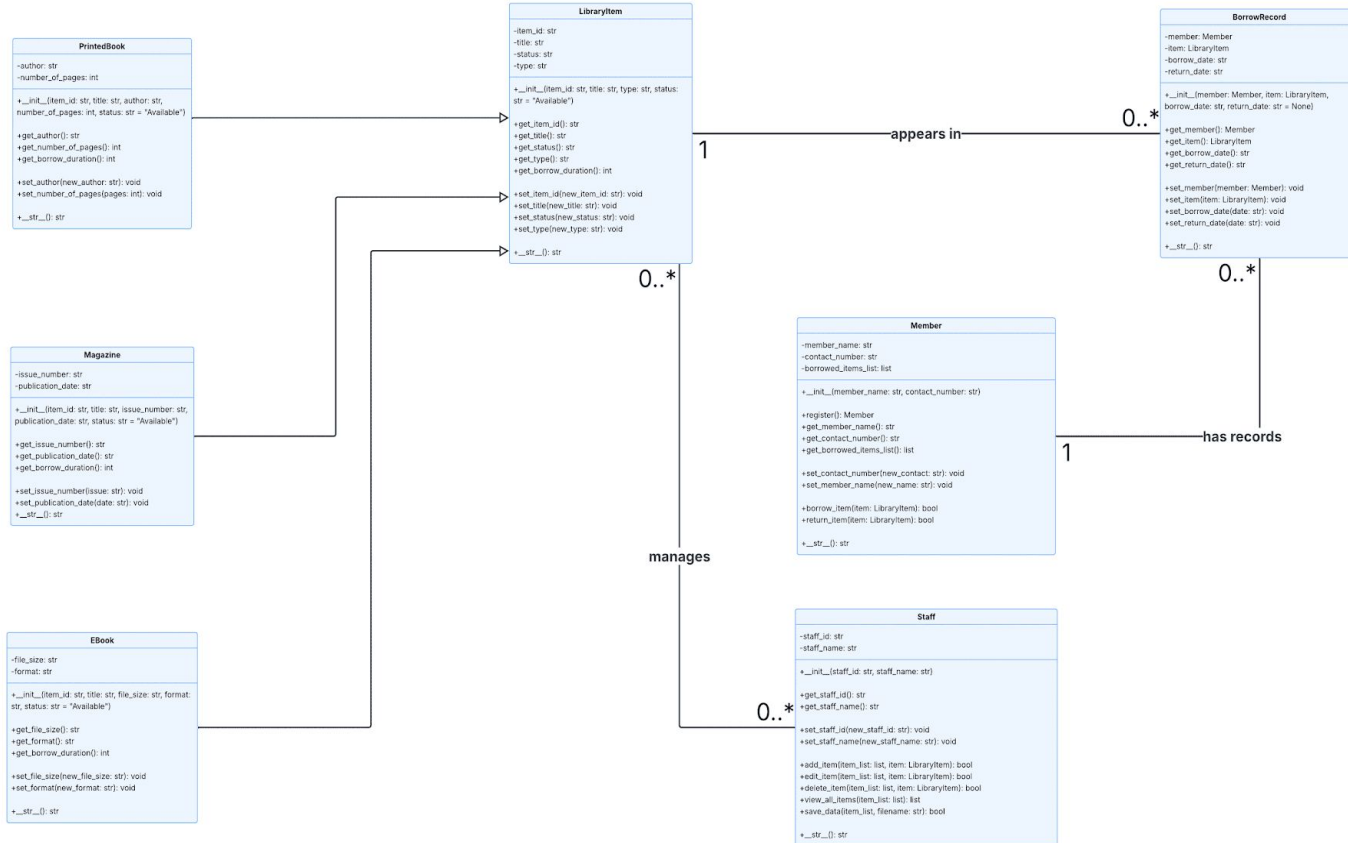
[https://lucid.app/lucidchart/324c341c-e4e8-44f1-908e-0c4d23fe29b6/edit?viewport\\_loc=1002%2C-1560%2C4178%2C1830%2C0\\_0&invitationId=in\\_v\\_bd26e3c6-e90c-446d-b0d6-c2513e5be6eb](https://lucid.app/lucidchart/324c341c-e4e8-44f1-908e-0c4d23fe29b6/edit?viewport_loc=1002%2C-1560%2C4178%2C1830%2C0_0&invitationId=in_v_bd26e3c6-e90c-446d-b0d6-c2513e5be6eb)

*If prompted by Lucidchart, please sign in with your account to view the interactive diagram.*

# Assignment 2 MILESTONE 1 (Team 3)

Members: Angad, Nathan, Dyllon, Alexander

UML diagram



## How is Inheritance and Polymorphism used?

Inheritance is used in the Smart Library Management System by making `LibraryItem` the base class for `PrintedBook`, `EBook`, and `Magazine`. The common attributes and methods, such as `item_id`, `title`, `status`, and `get_borrow_duration()`, can be shared by the three subclasses. Each subclass also has its own specific attributes, such as `author` and `number of pages` for `PrintedBook`, `file size` and `format` for `EBook`, and `issue number` and `publication date` for `Magazine`.

Polymorphism is applied through the `get_borrow_duration()` method. Although each subclass uses the same method, it overrides it to provide a different borrowing duration depending on the subclass itself. A `PrintedBook` can be borrowed for up to 14 days, an `EBook` for up to 7 days, and a `Magazine` for up to 3 days.

## AI Acknowledgement:

- ChatGPT was used to create text, suggestions, and assistance related to reviewing and developing class materials, including the UML itself. Prompts regarding the UML itself, diagrams, were used to assist with the related class design and content. Link: <https://chatgpt.com/share/6a898165-1a30-83ec-9ec9-cc33fcb1683c>.
- Claude was used to generate text, suggestions, assistance used in developing those materials itself. Claude was given prompts regarding the development of the classes, including refining the classes and relationships of it. Link: <https://claude.ai/share/79df42b8-4fd9-40d1-9dc8-c57e3b5d9021>.
- Gemini was used to generate text, suggestions with an emphasis on refining the classes. We used prompts requesting assistance with the design, development and reviewing the content of the classes. Link: <https://share.gemini.google/wqw57H7SkRyb>.